# SHRINKING THE OPEN SO THE SUPPORT OF THE DEGREE ZERO PART IS PROPER OVER THE BASE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: henselian level pairs, support identification after base change, the henselian open and closed split-off of the support, and the shrinking of W by the part disjoint from the special fibre

*Editorial note.* The label lemma-proper-support names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Henselianity of the pair (A, I_n) for I_n the kernel of A -> A_n
- The scheme theoretic support of a finitely presented module as a closed subspace
- The stage-wise identification of the support after base change to A_n from the finitely presented lemma of the situation
- The henselian split-off lemma writing a closed subspace as a disjoint union of an open and closed part proper over the base and a part disjoint from the closed fibre

*Source reference.* Original reference 0CTD: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2388–2388. Prerequisite pointer invoked at proof line(s) 2575; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0CTH: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2508–2508. Prerequisite pointer invoked at proof line(s) 2576; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0CT7: sources/stacks-chow/tags.txt, source lines 16057–16057. Prerequisite pointer invoked at proof line(s) 2583; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 08B7: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2362–2362. Prerequisite pointer invoked at proof line(s) 2574; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0CWI: sources/stacks-chow/tags.txt, source lines 16173–16173. Prerequisite pointer invoked at proof line(s) 2592; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In Situation 08B7 let $K$ be as in Lemma 0CTD. Let $W \subset X$ be as in Lemma 0CTH. Set $\mathcal{F} = H^0(K)|_W$. Then, after possibly shrinking the open $W$, the support of $\mathcal{F}$ is proper over $A$.*

*Proof.* Fix $n \geq 1$. Let $I_n = \mathrm{Ker}(A \to A_n)$. By More on Algebra, Lemma 0CT7 the pair $(A, I_n)$ is henselian. Let $Z \subset W$ be the scheme theoretic support of $\mathcal{F}$. This is a closed subspace as $\mathcal{F}$ is of finite presentation. By part (3) of Lemma 0CTH we see that $Z \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A_n)$ is equal to the support of $\mathcal{F}_n$ and hence proper over $\mathrm{Spec}(A/I)$. By More on Morphisms of Spaces, Lemma 0CWI we can write $Z = Z_1 \amalg Z_2$ with $Z_1, Z_2$ open and closed in $Z$, with $Z_1$ proper over $A$, and with $Z_1 \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/I_n)$ equal to the support of $\mathcal{F}_n$. In other words, $|Z_2|$ does not meet $X_0$. Hence after replacing $W$ by $W \setminus Z_2$ we obtain the lemma. $\square$